NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University*)*

Department of Chemistry

CHY104-MATERIALS CHEMISTRY FOR DEVICES AND E-WASTE MANAGEMENT

MULTIPLE CHOICE QUESTIONS

UNIT – II CORROSION SCIENCE & MULTIPLE FINISHING

1.	**Identify the incorrect statement/s.** **(i) Corrosion is a natural process initiating on the surface of the metal.** **(ii) Corrosion reduces metal to its ionic form.** **(iii) Rate of corrosion is independent of the environment.**			
	A)	(ii) and (iii)	B)	(i) and (ii)
	C)	Only (iii)	D)	Only (i)
Ans	**A)**	**(ii) and (iii)**		
2.	**Situation favorable for dry corrosion is ……..**			
	A)	Presence of moisture	B)	Presence of conducting electrolyte
	C)	Presence of oxygen or other gases	D)	Presence of non-polar liquids
Ans	**C)**	**Presence of oxygen or other gases**		
3.	**Reduction of water / moisture in presence of oxygen leads to the formation of ……..**			
	A)	Hydrogen gas only	B)	Hydroxyl ions only
	C)	Both hydrogen gas and hydroxyl ions	D)	Oxide anions
Ans	**B)**	**Hydroxyl ions only**		
4.	**Corrosion that takes place when two dissimilar metals are coupled and exposed to the same environment, is known as ……..**			
	A)	Galvanic corrosion	B)	Dry corrosion
	C)	Pitting corrosion	D)	Passivation
Ans	**A)**	**Galvanic corrosion**		
5.	**Which of the following statements holds true for Galvanic Corrosion?**			
	A)	More cathodic metal gets corroded.	B)	Less noble metal gets corroded.
	C)	The metal having a higher standard reduction potential gets corroded.	D)	The metal placed lower in the galvanic series get corroded.
Ans	**B)**	**Less noble metal gets corroded.**		
6.	**A steel tank fitted with a copper tap undergoes corrosion in water because ……**			
	A)	Electrode potential of both metals are almost equal	B)	Steel acts as an anode and corrodes due to galvanic action
	C)	Steel is susceptible to water	D)	Copper acts as an anode and corrodes
Ans	**B)**	**Steel acts as an anode and corrodes due to galvanic action**		
7.	**Complete the statement: In differential aeration corrosion, ……..**			
	A)	Oxygen-deficient part gets protected.	B)	Oxygen-deficient part acts as anode.
	C)	Oxygen-rich part acts as anode.	D)	Metal as a whole acts as cathode.
Ans	**B)**	**Oxygen-deficient part acts as anode**		
8.	**The accelerated corrosion of iron under defective tin coatings is due to ……**			

	A)	Change in electrode potential	B)	Acidic environment
	C)	Small cathodic surface area	D)	Small anodic surface area
Ans	**D)**	**Small anodic surface area**		
9.	**For the accelerated corrosion to take place ratio of anodic to cathodic area should be ……..**			
	A)	Greater than 1	B)	Lesser than 1
	C)	Equal to 1	D)	Greater than 1.5
Ans	**B)**	**Lesser than 1**		
10.	**The metal, which is protected by a layer of its own oxide, is ……..**			
	A)	Al	B)	Pt
	C)	Zn	D)	Fe
Ans	**A)**	**Al**		
11.	**In general, corrosion is maximum when the pH of the corroding medium is ……..**			
	A)	Above 7.0	B)	Equal to 7.0
	C)	Below 7.0	C)	Equal to 10.0
Ans	**C)**	**Below 7.0**		
12.	**Identify the correct statement/s:** **(i) Lower the hydrogen overvoltage, more is the rate of corrosion.** **(ii) Smaller the anodic area (larger cathodic area), more is the rate of corrosion.**			
	A)	Only (i)	B)	Only (ii)
	C)	Both (i) and (ii)	D)	Neither (i) nor (ii)
Ans	**C)**	**Both (i) and (ii)**		
13.	**Iron corrodes faster in ……..**			
	A)	Hard water	B)	Soft water
	C)	Demineralised water	D)	Distilled water
Ans	**A)**	**Hard water**		
14.	**CPR is expressed in ……. unit.**			
	A)	mg per year	B)	mm per year
	C)	mg per hr	D)	mm per hr
Ans	**B)**	**mm per year**		
15.	**What is the role of NH_4Cl flux during galvanization process?**			
	A)	Prevents oxidation of Zn	B)	Prevents oxidation of iron
	C)	Helps to maintain constant bath temperature	D)	Prevents chemical reaction between Zn and Fe
Ans	**A)**	**Prevents oxidation of Zn**		
16.	**In anodic coatings, the coating metals possess …….. reduction potential than base metal**.			
	A)	Higher	B)	medium
	C)	Lower	D)	equal
Ans	**C)**	**Lower**		
17.	**Anodic metal coating is a process of coating the base metal with ……..**			
	A)	more reactive metal	B)	inert metal
	C)	less reactive metal	D)	passive metal
Ans	**A)**	**more reactive metal**		
18.	**Chemical conversion coating is ……. type of coating.**			
	A)	Organic	B)	Inorganic
	C)	Polymer	D)	Alloy
Ans	**B)**	**Inorganic**		

19.	**Anodization of corrosion control technique is based on the principle of ……**			
	A)	Chemical coating	B)	Chemical conversion coating
	C)	Metal coating	D)	Electroless plating
Ans	**B)**	**Chemical conversion coating**		
20.	**Sacrificial anode method is widely used in ships and underground pipelines as ……**			
	A)	It is highly decorative.	B)	It ensures uniform strength of the material.
	C)	It continuously protects the base metal by corroding the anodic metal.	D)	It eliminates the use of anodic metal.
Ans	**C)**	**It continuously protects the base metal by corroding the anodic metal.**		
21.	**Sacrificial anodes used in cathodic protection of iron are generally made of ……**			
	A)	Copper, Nickel, Lead	B)	Magnesium, Zinc, Aluminium
	C)	Carbon, Platinum, Gold	D)	Iron, Chromium, Silver
Ans	**B)**	**Magnesium, Zinc, Aluminium**		
22.	**To protect the buried pipeline from corrosion, it is connected to Mg piece through a wire. This process is called as ……**			
	A)	Impressed current cathodic protection	B)	Galvanic protection
	C)	Sacrificial cathodic protection	D)	Sacrificial anodic protection
Ans	**D)**	**Sacrificial anodic protection**		
23.	**Which of the following is not a corrosion prevention technique?**			
	A)	Sacrificial Anode	B)	Protective coating
	C)	Pickling	D)	Anodizing
Ans	**C)**	**Pickling**		

24.	**In the electroplating process, generally the object to be plated is made as ……**			
	A)	Anode	B)	Electrolyte
	C)	Cathode	D)	Power house
Ans	**C)**	**Cathode**		
25.	**In electrodeposition process, complexing agents are added to ……**			
	A)	increase the chemical reaction between the cathode metal and plating ions	B)	make the potential of the plating ion more negative
	C)	enhance the passivation of anodes	D)	decrease the throwing power of the plating bath
Ans	**B)**	**make the potential of the plating ion more negative**		
26.	**The electrolyte commonly used for chromium electroplating is ……**			
	A)	H_2SO_4	B)	$CrO_3 + H_2SO_4$
	C)	CrO_3 + HCl	D)	HCl solution
Ans	**B)**	**$CrO_3 + H_2SO_4$**		
27.	**During chromium plating, …… is used as anode.**			
	A)	Chromium	B)	Nickel
	C)	Calomel electrode	D)	Inert electrode
Ans	**D)**	**Inert electrode**		
28.	**Chromium anodes are not used in Cr plating because**			
	A)	Chromium metal passivates strongly in acid sulphate medium.	B)	Chromium metal activates strongly in acid sulphate medium.
	C)	Chromium anode gives rise to Cr^{6+} ions on dissolution.	D)	Chromium metal oxidizes the base component

Ans	**A)**	**Chromium metal passivates strongly in acid sulphate medium.**		
29.	**Electroless plating is characterized by ……..**			
	A)	The use of an external electrical current.	B)	A chemical reduction process without external current.
	C)	Electrodeposition of metals.	D)	The need for a conductive substrate.
Ans	**B)**	**A chemical reduction process without external current.**		
30.	**Identify the advantage of electroless plating over electroplating technique.**			
	A)	Higher deposition rates.	B)	Better throwing power of the bath solution.
	C)	Lower cost.	D)	Easier to control the deposition process.
Ans	**B)**	**Better throwing power of the bath solution.**		
31.	**During the fabrication of PCB, the electroless plating of copper is performed using …… as reducing agent.**			
	A)	Sodium hypophosphate	B)	Sodium borohydride
	C)	Hydrazine	D)	Formaldehyde
Ans	**D)**	**Formaldehyde**		
32.	**In the production of double-sided printed circuit boards, connection-through-hole is efficiently made by ……..**			
	A)	Electroplating	B)	Electroless plating
	C)	Molten dip method	D)	using copper wires
Ans	**B)**	**Electroless plating**		
33.	**Which of the following statements is correct regarding the difference between electroless and electrolytic plating?**			
	A)	Electroless plating includes reduction reaction, while electrolytic plating does not.	B)	Electrolytic plating is a chemical process, while electroless plating is an electrochemical process.
	C)	Electroless plating offers more uniform coatings than electrolytic plating.	D)	Electrolytic plating is used for through-hole plating, while electroless plating is not
Ans	**C)**	**Electroless plating offers more uniform coatings than electrolytic plating.**		
34.	**Electroless plating is used in the electronics industry for ……..**			
	A)	Creating decorative finishes on electronic components.	B)	Increasing the electrical resistance of circuits.
	C)	Creating conductive pathways on printed circuit boards (PCBs).	D)	Providing corrosion protection for metal components.
Ans	**C)**	**Creating conductive pathways on printed circuit boards (PCBs).**		
35.	**In electronics, the abbreviation PCB stands for ……..**			
	A)	Polished Circuit Board	B)	Printed Circular Board
	C)	Polymer Circuit Board	D)	Printed Circuit Board
Ans	**D)**	**Printed Circuit Board**		

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